



School of Future Tech

Case Study Report

on

**Case Study 23. GigGuide: Independent Music
Ticketing**

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1. Introduction to the Case Study

Event discovery and ticket booking have become an important part of modern digital services. People often attend concerts, comedy shows, sports matches, festivals, and nightlife events, but finding and booking tickets for these events can sometimes be inconvenient or scattered across multiple platforms.

This case study focuses on designing and implementing a web-based event ticketing platform called "GigGuide." The platform, heavily inspired by platforms like District, allows users to explore events happening around them, filter events by category, and book tickets directly through a simple interface.

The main objective of the project is to build a lightweight event discovery and ticket booking system using HTML, CSS, and JavaScript. The platform includes features such as:

- Event browsing
- Category filtering
- Interactive event cards
- Ticket selection
- Quantity selector
- Dynamic price calculation
- Responsive design for mobile devices

The case study highlights how modern web technologies can be used to create a functional and visually appealing event ticketing interface.

2. Problem Statement

Many event platforms primarily focus on large concerts or mainstream events, leaving smaller or local events with limited exposure. Additionally, users often find it difficult to browse events efficiently or quickly compare ticket options. The objective of this project is to design a simple event ticketing platform focused on usability and event discovery.

This project aims to provide a clean, simple and intuitive user experience for event browsing and ticket purchasing.

3. Case Study Design

The system is designed as a two-page web application.

Page 1: Events Page

This page acts as the main landing page where users can browse all available events.

Features include:

- Navigation bar with platform branding
- Spotlight slider displaying featured events
- Event categories for filtering
- Dynamic grid showing event cards
- Ticket booking links

Each event card contains:

- Event image
- Event name
- Venue
- Date
- Ticket price
- "Book Ticket" button

Clicking the Book Ticket button redirects the user to the ticket purchasing page.

Page 2: Ticket Purchase Page

This page allows the user to:

- View event details
- Select ticket type
- Choose ticket quantity
- See the total price dynamically calculated

The page retrieves the event ID from the URL and loads the correct event information using JavaScript.

System Architecture

The application consists of the following components:

- **HTML** – Structure of the web pages
- **CSS** – Styling and responsive design
- **JavaScript** – Interactive features and logic

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

HTML

HTML is used to structure the layout of the website, including:

- Navigation bar

- Event cards
- Category filters
- Ticket booking interface

Two main HTML pages were created:

- events.html – Displays the list of events
- tickets.html – Handles ticket booking

CSS

CSS is used to design and style the website.

Major design features include:

- Modern gradient backgrounds
- Glassmorphism navbar
- Event card hover animations
- Responsive layouts
- Mobile-first design
- Styled ticket purchase interface

CSS also ensures the website adapts to different screen sizes using media queries.

JavaScript

JavaScript provides the interactive functionality of the platform.

Main JavaScript features include:

Event Data Management:

All event information is stored in a JavaScript array of objects, including:

- Event ID
- Name
- Category
- Venue
- Date
- Ticket prices
- Event image

Event Rendering:

JavaScript dynamically generates event cards on the page using DOM manipulation.

Category Filtering:

Users can filter events by category such as:

- Music
- Nightlife
- Comedy
- Sports
- Festivals

The filtering system updates the displayed events dynamically.

Image Slider:

A spotlight image slider is implemented using JavaScript to showcase featured events with navigation buttons.

Ticket Quantity Selector:

Users can increase or decrease ticket quantity using buttons.

Price Calculation:

The total ticket price is automatically calculated using the formula:

Total Price = Ticket Price × Quantity

This calculation updates dynamically whenever the ticket type or quantity changes.

5. Problem Statement / Case Study Implementation Details and Snapshots.

Navigation Bar:

- Platform logo
- User location display
- Profile icon

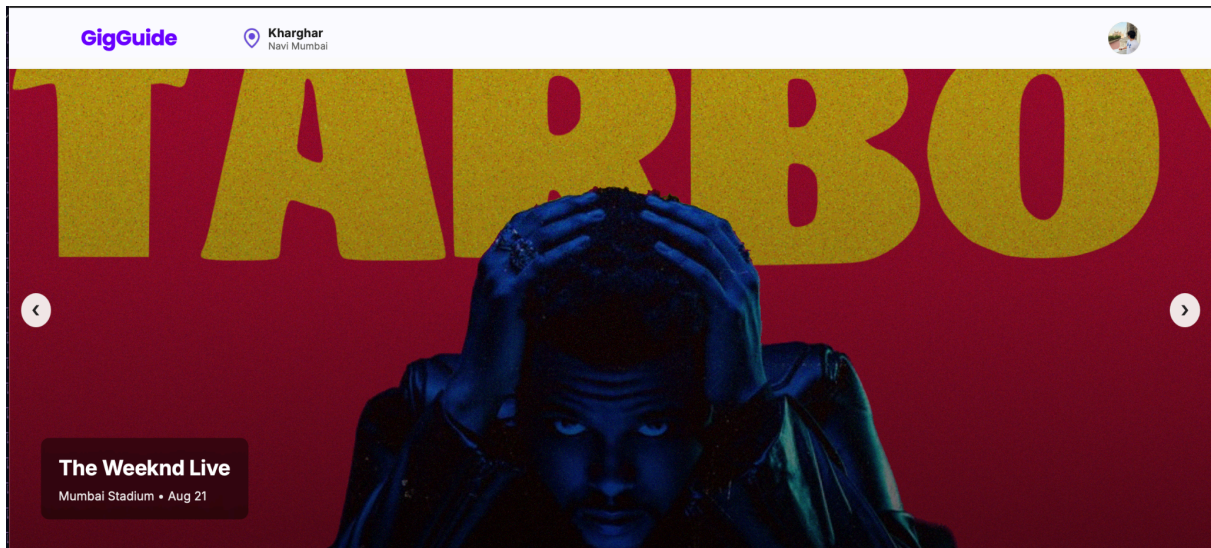
It remains sticky at the top of the screen for better usability.

Event Slider:

A large slider highlights featured events such as:

- The Weeknd Live
- Rolling Loud India
- IPL Matches
- Champions League Final
- Dua Lipa Live

The slider allows users to navigate through featured events.



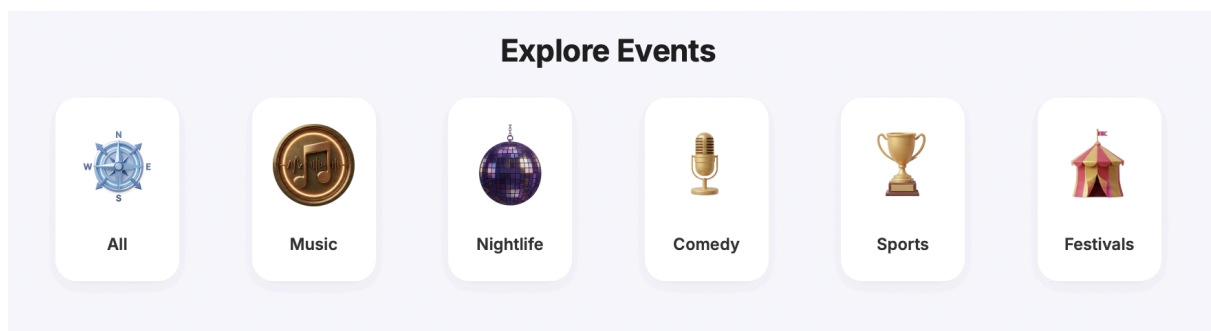
Category Section:

Users can explore events through different categories.

Categories implemented include:

- All Events
- Music
- Nightlife
- Comedy
- Sports
- Festivals

Clicking a category dynamically filters the event list.



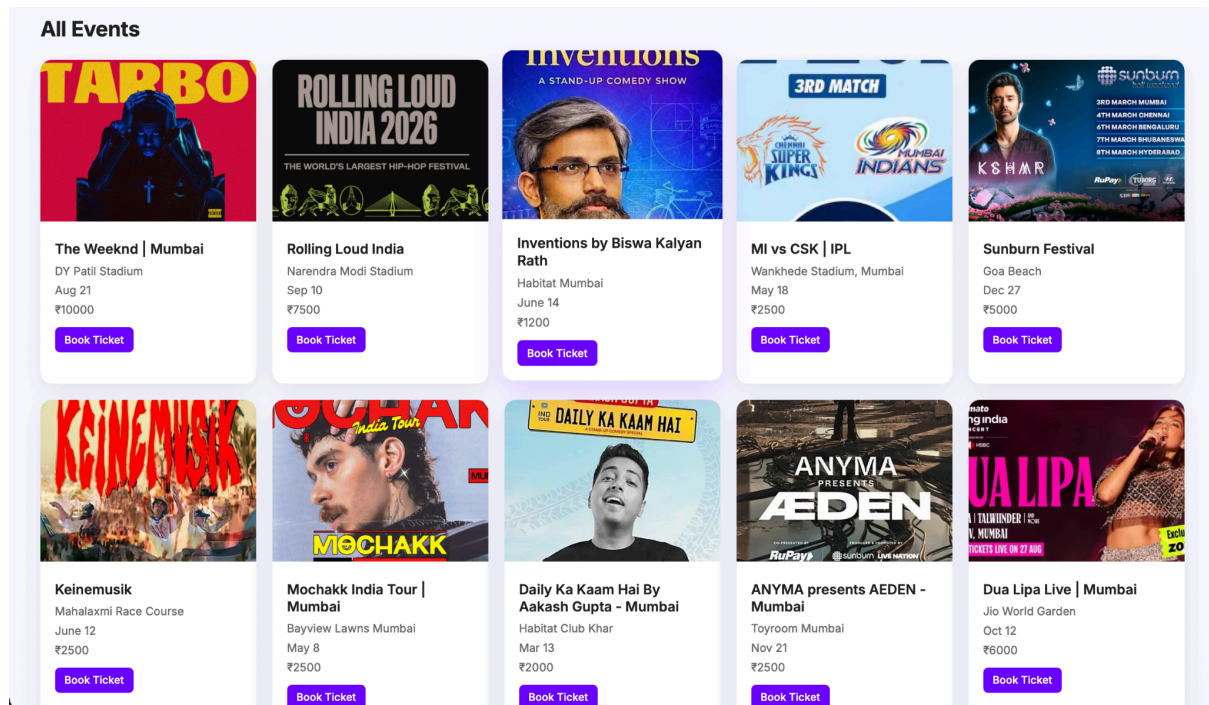
Event Cards:

Each event card displays:

- Event image
- Event title
- Venue
- Event date

- Ticket price
- Booking button

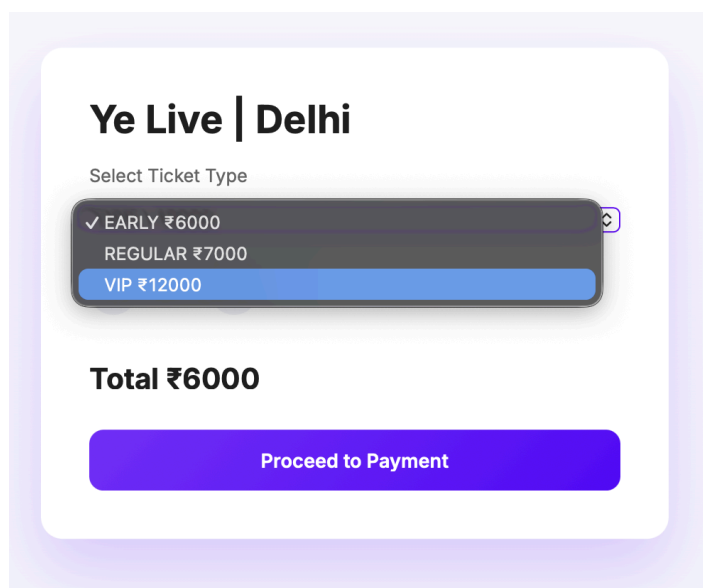
Hover animations improve visual interaction.



Ticket Booking Page:

The ticket purchase page allows users to:

- Select ticket type (Early Bird, Regular, VIP)
- Adjust ticket quantity
- View total ticket price



6. Problem Statement / Case Study Results and Conclusion.

The GigGuide event ticketing platform successfully demonstrates how modern web technologies can be used to create an interactive event discovery and ticket booking system.

The project achieved its objectives by implementing:

- Dynamic event listing
- Category filtering
- Interactive event cards
- Ticket booking interface
- Dynamic price calculation
- Responsive design

The system provides a clean, simple and intuitive user experience, making it easy for users to explore and book events.

Although the platform currently works as a frontend prototype, future improvements could include:

- Payment gateway integration
- Location services integration
- User authentication
- Event organizer dashboards
- Actual database integration
- Real-time ticket availability

Overall, the project demonstrates how HTML, CSS, and JavaScript can be used to build functional and visually appealing web applications.

7. References

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- Google Fonts Documentation
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